IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE/EIE PART IV: MEng and ACGI

DIGITAL IMAGE PROCESSING

Wednesday, 6 May 2026, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer THREE questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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1. (a) (i) I have a digital image $f(x,y)$ of size $N \times N$. If I define a new image $g(x,y)$ by transposing the original, i.e., $g(x,y) = f(y,x)$, what is the two-dimensional Discrete Fourier Transform (2D DFT) of $g(x,y)$ in terms of the 2D DFT of $f(x,y)$? **[4]**

(ii) I have a digital image $f(x,y)$ of size $N \times N$. If I define a new image $g(x,y)$, such as $g(x,y) = f(N-x, N-y)$, what is the two-dimensional Discrete Fourier Transform (2D DFT) of $g(x,y)$ in terms of the 2D DFT of $f(x,y)$? **[4]**

For the purposes of applying DFTs, all images are treated as if they were periodic with period N in both spatial dimensions.

(b) Consider the population of vectors $\underline{f}$ of the form

$$\underline{f}(x,y) = \begin{bmatrix} f_0(x,y) \\ f_1(x,y) \\ \vdots \\ f_{N-1}(x,y) \end{bmatrix}.$$

Each component $f_i(x,y),\ i = 0, 1, \ldots, N-1$ represents a real image of size $M \times M$. The population arises from the formation of the vectors $\underline{f}$ across the entire collection of pixels (x,y). Consider now a population of vectors $\underline{g}$ of the form:

$$\underline{g}(x,y) = \begin{bmatrix} g_0(x,y) \\ g_1(x,y) \\ \vdots \\ g_{N-1}(x,y) \end{bmatrix}$$

where the vectors $\underline{g}$ are the Karhunen-Loeve (KL) transforms of the vectors $\underline{f}$.
The resulting images are uncorrelated, and their variances are assumed to decay exponentially as follows. Variances have been normalised such that the maximum variance equals 1.

$$\text{Variance}\{g_i(x,y)\} = a^i, i = 0, \cdots, N-1, 0 < a < 1$$

You wish to retain only the first $K, K \leq N$ KL transform components such that the total energy retained is at least 25%. Determine the smallest integer K that satisfies this condition, in terms of a and N. Re-express the result if $N \to \infty$. **[6]**
[To solve this, you will need to use the geometric sum formula $\sum_{i=0}^{N-1} a^i = \frac{1-a^N}{1-a}, a \neq 1$.
You will also need the ceiling function $\lceil \cdot \rceil$, which rounds any number up to the nearest integer.]

(c) I have a digital image $f(x,y)$ of size $N \times N$. The image $g(x,y)$ is the so-called even-symmetric extension of $f(x,y)$ defined as follows:

$$g(x,y) = \begin{cases} f(x,y) & 0 \leq x < N, 0 \leq y < N \\ f(2N-1-x,y) & N \leq x < 2N, 0 \leq y < N \\ f(x,2N-1-y) & 0 \leq x < N, N \leq y < 2N \\ f(2N-1-x,2N-1-y) & N \leq x < 2N, N \leq y < 2N \end{cases}$$

It can be shown that the 2D DFT of $g(x,y)$ produces the same frequency coefficients (up to scaling) as the 2D DCT of the original image $f(x,y)$. Based on this fact, explain why the DCT exhibits excellent compaction properties. **[6]**

2. (a) (i) Knowing that adding uncorrelated images results in the convolution of their histograms, how would you expect the contrast of the sum of two uncorrelated images to compare with the contrast of its component images? Justify your answer. **[4]**

(ii) Consider an $N \times N$ image $f(x, y)$. From $f(x, y)$ create an image

$$g(x, y) = 4f(x, y) - f(x, y - 1) - f(x, y + 1) - f(x - 1, y) - f(x + 1, y)$$

Comment on the histogram of $g(x, y)$ in relation to the histogram of $f(x, y)$. **[4]**

(b) Explain why additive white noise is generally more noticeable in smooth or slowly varying regions of an image than in regions with sharp edges or textures. Use both spatial-domain and frequency-domain reasoning in your answer. **[4]**

(c) Describe a method for implementing a bandpass filter in the spatial domain using convolution masks. Justify the design and explain how the method achieves bandpass behaviour. **[4]**

(d) Show that the Laplacian mask provides a discrete approximation to the local second-order derivative of an image. **[4]**

3. We are given the degraded version $g(x,y)$ of a discrete image $f(x,y)$ such that in lexicographic ordering

$$g = Hf + n \quad (\mathbf{3.1})$$

where H is the degradation matrix which is assumed to be block-circulant and n is the noise term which is assumed to be zero-mean, white and independent of the image f. The symbols f, n, g denote the lexicographically ordered original, noise and degraded image respectively. All images involved have size $N \times N$ after extension and zero-padding, with N an even integer.

(a) A particular image $f(x,y)$ is distorted by convolution with the spatially invariant point spread function $h(x,y)$ where:

$$h(x,y) = \frac{1}{8}[4\delta(x,y) + \delta(x-1,y) + \delta(x+1,y) + \delta(x,y-1) + \delta(x,y+1)]$$

Show that for any even N there may exist at least one discrete frequency pair (u,v) for which Inverse Filtering cannot be applied. Consequently, demonstrate how the method of Pseudoinverse Filtering for image restoration can be applied in this scenario. **[4]**

(b) The Constrained Least Squares (CLS) class of filters is designed to address potential issues that may arise during the application of Inverse Filtering to restore image f from **Equation (3.1)** above. Their mathematical formulation is given below:

$$\frac{H(u,v)^*}{|H(u,v)|^2 + \alpha|C(u,v)|^2} \quad (\mathbf{3.2})$$

where α is the regularization parameter and $C(u,v)$ represents the Laplacian operator. Identify the frequency pair (u,v) for which the 2D DFT $H(u,v)$ must be non-zero in order to recover the original image using **Equation (3.2)** above. **[4]**

(c) Consider a Wiener filter of the form:

$$W(u,v) = \frac{H(u,v)^*}{|H(u,v)|^2 + 1/K} \quad (\mathbf{3.3})$$

where K is a positive constant and $K \neq 0$.

(i) Derive the form of **Equation (3.3)** above from the general Wiener filter equation assuming constant signal-to-noise ratio. **[4]**

(ii) Discuss the effect of the parameter K on the trade-off between noise suppression and detail preservation. What happens to the restored image as $K \to 0$ and as $K \to +\infty$? **[4]**

(iii) For a given degraded image, suggest a theoretically sound way to estimate K, and justify your approach. **[4]**

4. (a) A 3 −level quantised image is to be Huffman-coded in blocks of M pixels. The successive pixel values are independent from each other. We assume that the pixel values are normalized, meaning that the maximum pixel value is 1.

intensity	percentage of pixels
1/6	10
3/6	20
5/6	70

What are the Huffman codes for:

(i) $M = 1$ **[2]**

(ii) $M = 2$ **[3]**

(iii) For each of the above two cases, calculate the coding efficiency and the redundancy of each codebook. **[2 points for each M; total 4 points]**

(b) A Discrete Memoryless Source (DMS) emits the four symbols, **C**, **A**, **T**, and **S** with the following probabilities:

symbol	probability
C	1/8
A	1/8
T	1/4
S	1/2

The encoded symbols are transmitted over a binary channel with the following characteristics:

- A transmitted bit '0' is received as '1' with probability 0.01
- A transmitted bit '1' is received as '0' with probability 0.10

(i) Construct the Huffman code for the given source that minimises the probability of error over the specified channel. Clearly indicate the binary codewords assigned to each symbol and compute the average codeword length. **[3]**

(ii) Compute the entropy of the source (in bits/symbol). Compare it with the average length of your Huffman code and comment on the efficiency. **[2]**

(iii) Assume the word "**CAT**" is encoded using your Huffman code and transmitted through the channel described above. Using the bit error probabilities, estimate the probability that the entire word "**CAT**" is received incorrectly. You may assume bit errors are independent and that a word is received incorrectly if at least one bit is corrupted. **[6]**